Michael Evans

mevansci.github.io

mevan028@odu.edu (757) 236-1257

EDUCATION

Old Dominion University Norfolk, VA
M.S. in Electrical & Computer Engineering Expected Aug 2027
Thesis: *Multimodal Disease-Agnostic Regional Brain Glucose Hypometabolism Estimation*
GPA: 3.68/4.00

B.S. in Computer Science May 2025
GPA: 3.85/4.00

Tidewater Community College Norfolk, VA
A.S. in Computer Science May 2023

RESEARCH INTERESTS

Deep learning, medical image analysis, neuroimaging, multimodal learning, neurodegenerative disease modeling

RESEARCH EXPERIENCE

Graduate Research Assistant Sept 2025-Present
Old Dominion University, advised by Dr. Khan Iftekharuddin
Vision Lab

- Trained a multimodal CNN-Transformer model from scratch to predict regional age-adjusted FDG-PET uptake, enabling hypometabolic profile prediction from MRI scans and CSF biomarkers.
- Developed a corrective diffusion alternative to image harmonization to reduce sensitivity to training site and scanner stratification for brain tumor segmentation by up to 32.1% across three multi-site MRI datasets.
- Designed a standardized, BIDS-compliant preprocessing pipeline for multimodal neuroimaging (sMRI, fMRI, FDG-PET) using MATLAB, fMRIPrep, and Clinica, ensuring reproducible research outcomes.

Undergraduate Research Assistant Sept 2024-Sept 2025
Old Dominion University, advised by Dr. Khan Iftekharuddin
Vision Lab

- Trained a random forest classifier for atypical Alzheimer's disease classification using cognitive test scores and MRI features, improving recall from 34% to 77% on ADNI and 52% to 69% on NACC over baseline.
- Constructed a FreeSurfer pipeline on an HPC cluster to extract regional volumetric MRI features at scale.
- Visualized an ROI-map on an anatomically averaged brain template to highlight brain regions with significant features ($\alpha = 0.05$) for Alzheimer's classification with volumetric group mean z-score color coding.

NSF REU Intern - Developing Self-Drive Algorithms May-July 2024
Lawrence Tech and Michigan State University, advised by Drs. Chan-Jin Chung and Josh Siegel
Computer Science & Artificial Intelligence Robotics Lab

- Designed an adaptive speed algorithm to reduce acceleration and braking in autonomous vehicles through intersections by up to 75.35%, minimizing fuel consumption and noise pollution.
- Implemented and analyzed DBSCAN and K-means machine learning self-drive algorithms for reliability.
- Built a simulation in GazelleSim for testing our autonomous intersection control in a controlled environment.

NSF REU Intern - Disinformation Detection and Analytics June-Aug 2023
Old Dominion University, advised by Dr. Jian Wu
Lab for Applied Machine Learning and Natural Language Processing Systems

- Engineered prompts and tuned hyperparameters with the OpenAI API for scientific claim verification.
- Increased the size of our project corpus by over 200% to improve benchmarking the effectiveness of the GPT-3.5-turbo model's ability to generalize across multiple scientific news domains.
- Calculated the precision, recall, and F1 score on zero-shot classification with the GPT-3.5-turbo model against our dataset on two sub-tasks: stance labeling and identifying sentence rationales.

PUBLICATIONS

Preprints and Manuscripts Under Review

[**2**] M. L. Evans, M. F. Bin Hossen, M. S. Sadique, W. Farzana, and K. M. Iftekharuddin. A Prediction-Level Alternative to Image Harmonization for Multi-Site Brain Tumor Segmentation. *Under review at Nature Scientific Reports.*

[**1**] M. A. Witherow, M. L. Evans, A. Temtam, H. R. Okhravi, and K. M. Iftekharuddin. Machine Learning-Enhanced Non-Amnestic Alzheimer's Disease Diagnosis From MRI and Clinical Features. *Under review at the Journal of Alzheimer's Disease.*

Peer-Reviewed Publications

[**9**] M. L. Evans, M. A. Witherow, M. F. Bin Hossen, and K. M. Iftekharuddin. Multimodal Estimation of Regional Hypometabolism from Structural MRI and CSF Biomarkers in Alzheimer's Disease. *Accepted at SPIE Medical Imaging 2027.*

[**8**] A. Sanda, I. Wann, M. S. Sadique, M. L. Evans, K. M. Iftekharuddin, and A. Musto. Utilizing Quantitative MRI Features to Characterize Seizure Status in Glioblastoma Patients. *Accepted at SPIE Medical Imaging 2027.*

[**7**] M. F. Bin Hossen, M. L. Evans, M. S. Sadique, W. Farzana, A. Rahman, A. Temtam, and K. M. Iftekharuddin. An SSIM-Prioritized 3D Attention U-Net for Synthetic Brain Inpainting. *Brain Tumor Segmentation (BraTS) Challenge at MICCAI*, 2026.

[**6**] M. F. Bin Hossen, M. L. Evans, M. S. Sadique, W. Farzana, A. Rahman, A. Temtam, and K. M. Iftekharuddin. Pediatric Brain Tumor Segmentation with MedSwinNet. *Brain Tumor Segmentation (BraTS) Challenge at MICCAI*, 2026.

[**5**] M. S. Sadique, M. F. Bin Hossen, M. L. Evans, W. Farzana, A. Rahman, A. Temtam, and K. M. Iftekharuddin. A Systematic Evaluation of Learning Strategies for Generalizable Brain Tumor Segmentation Across Heterogeneous Tumor Populations. *BraTS-GoAT: Segmentation Generalizability of Brain Tumor Sub-regions Across Tumors in Pre-operative MRI*, 2026.

[**4**] M. S. Sadique, M. F. Bin Hossen, M. L. Evans, W. Farzana, A. Rahman, A. Temtam, and K. M. Iftekharuddin. CarveMix-RC: Addressing Rare-Class Imbalance Through Lesion-Aware Synthetic Augmentation for Brain Metastasis Segmentation. *Brain Tumor Segmentation (BraTS) Challenge at MICCAI*, 2026.

[**3**] M. L. Evans, M. Machado, R. Johnson, L. Escamilla, A. Vadella, B. Froemming-Aldanondo, T. Rastoskueva, M. Jostes, D. Butani, R. Kaddis, C. Chung, J. Siegel. A Roadside Unit for Infrastructure Assisted Intersection Control of Autonomous Vehicles. *IEEE International Conference on Electro/Information Technology (EIT)*, 2025.

[**2**] B. Froemming-Aldanondo, T. Rastoskueva, M. L. Evans, M. Machado, A. Vadella, L. Escamilla, R. Johnson, M. Jostes, D. Butani, R. Kaddis, C. Chung, and J. Siegel. Evaluating Low-Resource Lane Following Algorithms for Compute-Constrained Automated Vehicles. *IEEE International Conference on Artificial Intelligence, Robotics, and Control (AIRC)*, 2025.

[**1**] M. L. Evans, D. Soós, E. Landers, and J. Wu. MSVEC: A Multidomain Testing Dataset for Scientific Claim Verification. *National Workshop for REU Research in Networking and Systems (REUNS @ MobiHoc)*, 2023.

SELECTED TECHNOLOGIES

Libraries & Frameworks
PyTorch ∘ OpenCV ∘ Pandas ∘ NumPy ∘ MONAI ∘ SPM ∘ FSL ∘ Nilearn ∘ FreeSurfer ∘ fMRIPrep ∘ Clinica

AWARDS & PROFESSIONAL SERVICE

1st place official leaderboard winner – MICCAI 2026 Challenge BraTS-GoAT	Oct 2026
Reviewer – MICCAI 2026 Challenge BraTS-Inpainting and BraTS-PEDs	Aug 2026
NSF CISE REU Travel Grant – $1,200	May 2025
Research Profile Highlight – Old Dominion University College of Sciences Newsletter	Nov 2024
BIO, GEO, CISE and OCE REU Travel Grant – $1,200	Sept 2023